WORKED SOLUTION

STEP III Specimen paper, Question 2

Pure • Binomial Inequalities and Limits

STEP III Specimen paper, Question 2

Read the question before you go on. It is in the STEP Support Programme's question database:

step.maths.org/questions/spec-s3-q2

The key idea

Writing $y = 1 + x$ with $x > 0$ turns powers of y into binomial expansions, and every term of such an expansion is non-negative. So we may throw away all but one term and still have a lower bound. Keeping the term in x gives a bound that grows like n, which beats any fixed K; to beat K after dividing by n we need a bound that grows like n^2, so we keep the term in x^2 instead.

Solution

The inequality

Let $x \geq 0$ and let n be a positive integer. By the binomial theorem,

$$(1+x)^n = 1 + nx + \binom{n}{2}x^2 + \cdots + x^n.$$

Since $x \geq 0$, every term after the first two is non-negative, so

$$(1+x)^n \geq 1 + nx > nx.$$

The final inequality is strict because of the 1, and that is what makes the result hold even when $x = 0$ (where it reads $1 > 0$).

Deducing that y^n eventually exceeds K

Let $y > 1$ and put $x = y - 1$, so $x > 0$. The first part gives, for every positive integer n,

$$y^n = (1+x)^n > n(y-1).$$

If $K \leq 0$ there is nothing to do: $y^n > 0 \geq K$ for every n, so $N = 1$ works. If $K > 0$, choose an integer N with

$$N > \frac{K}{y-1},$$

which we can do because $y - 1 > 0$, so the right-hand side is a fixed number. Then for every integer $n \geq N$,

$$y^n > n(y-1) \geq N(y-1) > K,$$

using $y - 1 > 0$ in the middle step. This is exactly what we needed. Note that the same N works for *all* $n \geq N$, not just for $n = N$, because the lower bound $n(y-1)$ increases with n.

Showing that y^n/n eventually exceeds K

The bound $y^n > n(y-1)$ only gives $y^n/n > y - 1$, which is a fixed number and is no use against a large K. We need a lower bound for y^n that grows faster than n.

Ask yourself Which term of the binomial expansion should we keep this time?

Keep the x^2 term. With $x = y - 1 > 0$, all terms of the expansion are positive, so for every positive integer n,

$$y^n = (1+x)^n \geq 1 + nx + \frac{n(n-1)}{2}x^2 > \frac{n(n-1)}{2}x^2.$$

(For $n = 1$ there is no x^2 term, but the final inequality still holds since its right-hand side is 0.) Dividing by $n > 0$,

$$\frac{y^n}{n} > \frac{(n-1)(y-1)^2}{2}.$$

The right-hand side now grows without limit as n increases, and that is what we need. If $K \leq 0$, then $N = 1$ works since $y^n/n > 0$. If $K > 0$, choose an integer N with

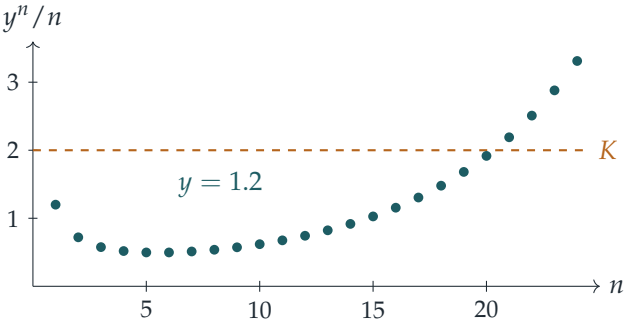
$$N > 1 + \frac{2K}{(y-1)^2}.$$

Then for every integer $n \geq N$ we have $n - 1 \geq N - 1 > 2K/(y-1)^2$, and so

$$\frac{y^n}{n} > \frac{(n-1)(y-1)^2}{2} > \frac{2K}{(y-1)^2} \cdot \frac{(y-1)^2}{2} = K,$$

which is the required result.

Again the bound increases with n, so the argument works for every $n \geq N$. This matters here: y^n/n itself is *not* always increasing. For y close to 1 it falls at first (for example, $y^2/2 < y$ whenever $y < 2$), as the sketch for $y = 1.2$ shows. So we could not simply find one n with $y^n/n > K$ and argue that later terms are larger.



Checks

Try $y = 1.01$ and $K = 1000$. Our recipes give $N = 100\,001$ for $y^n > K$ and $N = 20\,000\,002$ for $y^n/n > K$. A direct computation shows that $1.01^n/n > 1000$ already holds for every $n \geq 1424$. Our N is far from the smallest possible, but the question only asks for *some* N. We also tested both inequalities exactly with rational x for thousands of random x and n, and tested the chosen N (and many larger n) for thousands of random $y > 1$ and K.

Common slips

- Showing only that $y^N > K$ for one N, without explaining why $y^n > K$ for *every* $n \geq N$.
- Using $y^n > n(y-1)$ for the last part: dividing by n leaves a constant, which cannot beat an arbitrary K.
- Forgetting that K may be negative or zero, in which case $K/(y-1)$ is not a sensible lower bound on its own (though any N then works).

Beyond STEP

This is the first rigorous "ε–N" style argument most students meet: it proves that $y^n \to \infty$ and $y^n/n \to \infty$ when $y > 1$. The same trick, keeping the x^{k+1} term, shows that $y^n/n^k \to \infty$ for any fixed k, so exponentials beat every polynomial. Abbott's *Understanding Analysis* builds the theory of sequences and limits from exactly this kind of reasoning.

Beyond STEP worked solutions: https://beyondstep.keithmcnulty.org/. Solution text CC0 1.0; the question itself is © Cambridge University Press & Assessment and is not reproduced here.